OOP LAB Task 3

Q1. Create an array of String variables and initialize the array with the names of the months from January to December.

Code:

```
public class Main {
    public static void main(String[] args) {

        String[] months = {
            "January", "February", "March", "April",
            "May", "June", "July", "August",
            "September", "October", "November", "December"
        };

        for (int i = 0; i < months.length; i++) {
            System.out.println(months[i]);
        }
    }
}
```

Output:

```
Output
January
February
March
April
May
June
July
August
September
October
November
December

=== Code Execution Successful ===
```

Q2. Create an array containing 12 random decimal values between 0.0 and 100.0. Display the names of each month along with the corresponding decimal value. Calculate and display the average of the 12 values.

Code:

```
import java.util.Random;
```

```java
public class Main {
    public static void main(String[] args) {

        String[] months = {
            "January", "February", "March", "April",
            "May", "June", "July", "August",
            "September", "October", "November", "December"
        };

        double[] values = new double[12];

        Random rand = new Random();
        double sum = 0;

        for (int i = 0; i < values.length; i++) {
            values[i] = rand.nextDouble() * 100;
            sum += values[i];
        }

        System.out.println("Month\t\tValue");
        for (int i = 0; i < months.length; i++) {
            System.out.printf("%-10s\t%.2f\n", months[i], values[i]);
        }

        double average = sum / values.length;
        System.out.printf("\nAverage Value: %.2f\n", average);
    }
}
```

Output:

```
Output

Month       Value
January     83.20
February    91.54
March       69.14
April       42.82
May         85.72
June        65.73
July        48.37
August      43.40
September   1.84
October     39.34
November    40.07
December    60.23

Average Value: 55.95

=== Code Execution Successful ===
```

Q3. Write a Java program to calculate the average value of array elements.

Code:

```
public class Main {
    public static void main(String[] args) {

        int[] numbers = {2020, 2021, 2022, 2023, 2024, 2025, 2026};

        int sum = 0;

        for (int i = 0; i < numbers.length; i++) {
            sum += numbers[i];
        }
        double average = (double) sum / numbers.length;

        System.out.println("Average value: " + average);
    }
}
```

Output:

```
Output
Average value: 2023.0

=== Code Execution Successful ===
```

Q4. Write a Java program to reverse an array of integer values.

Code:

```
public class Main {
    public static void main(String[] args) {

        int[] numbers = {2020, 2021, 2022, 2023, 2024, 2025, 2026};

        System.out.println("Original Array:"); for
        (int i = 0; i < numbers.length; i++) {
            System.out.print(numbers[i] + " ");
        }

        int start = 0;
        int end = numbers.length - 1;

        while (start < end) {
            int temp = numbers[start];
            numbers[start] = numbers[end];
            numbers[end] = temp;

            start++;
            end--;
        }

         System.out.println("\nReversed Array:");
          for (int i = 0; i < numbers.length; i++) {
```

```
    } System.out.print(numbers[i] + " ");
  }
}
```

Output:

```
Output
Original Array:
2020 2021 2022 2023 2024 2025 2026
Reversed Array:
2026 2025 2024 2023 2022 2021 2020
=== Code Execution Successful ===
```

Q5. Write a Java program to sort a numeric array and a string array.

Code:

```
import java.util.Arrays;

public class Main {
public static void main(String[] args) { int[] numbers = {50, 20, 40, 10, 30};
String[] names = {"Babar", "Virat", "Rafay", "Mustafa", "Taha"};

Arrays.sort(numbers); Arrays.sort(names);

System.out.println("Sorted Numeric Array:"); for (int num : numbers) {
System.out.print(num + " ");
}

System.out.println("\n\nSorted String Array:"); for (String name : names) {
System.out.print(name + " ");
}
}
}
```

Output:

```
Output
Sorted Numeric Array:
10 20 30 40 50

Sorted String Array:
Babar Mustafa Rafay Taha Virat
=== Code Execution Successful ===
```

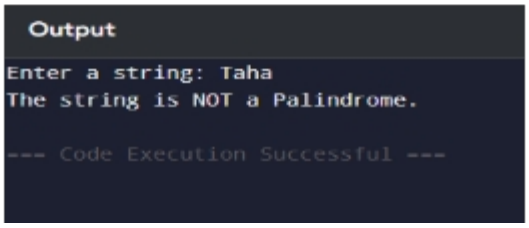
Q6. Write a Java program to check the given String is palindrome or not.

Code:

```
import java.util.Scanner;

public class Main {
```

```
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String original = str;
        String reversed = "";

        for (int i = str.length() - 1; i >= 0; i--) {
            reversed += str.charAt(i);
        }

        if (original.equalsIgnoreCase(reversed)) {
            System.out.println("The string is a Palindrome.");
        } else {
            System.out.println("The string is NOT a Palindrome.");
        }

        sc.close();
    }
}
```

Output:



Q7. Write a Java program to reverse words in a String given by user input.

Code:

```
import java.util.Scanner;

public class ReverseWords {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a sentence: ");
        String sentence = sc.nextLine();

        String[] words = sentence.split(" ");

        System.out.println("Reversed sentence:");
```

```
        for (int i = words.length - 1; i >= 0; i--) {
            System.out.print(words[i] + " ");
        }


        sc.close();
    }
}
```

Output:

```
Output
Enter a sentence: My Name is Ansaf
Reversed sentence:
Ansaf is Name My
=== Code Execution Successful ===
```